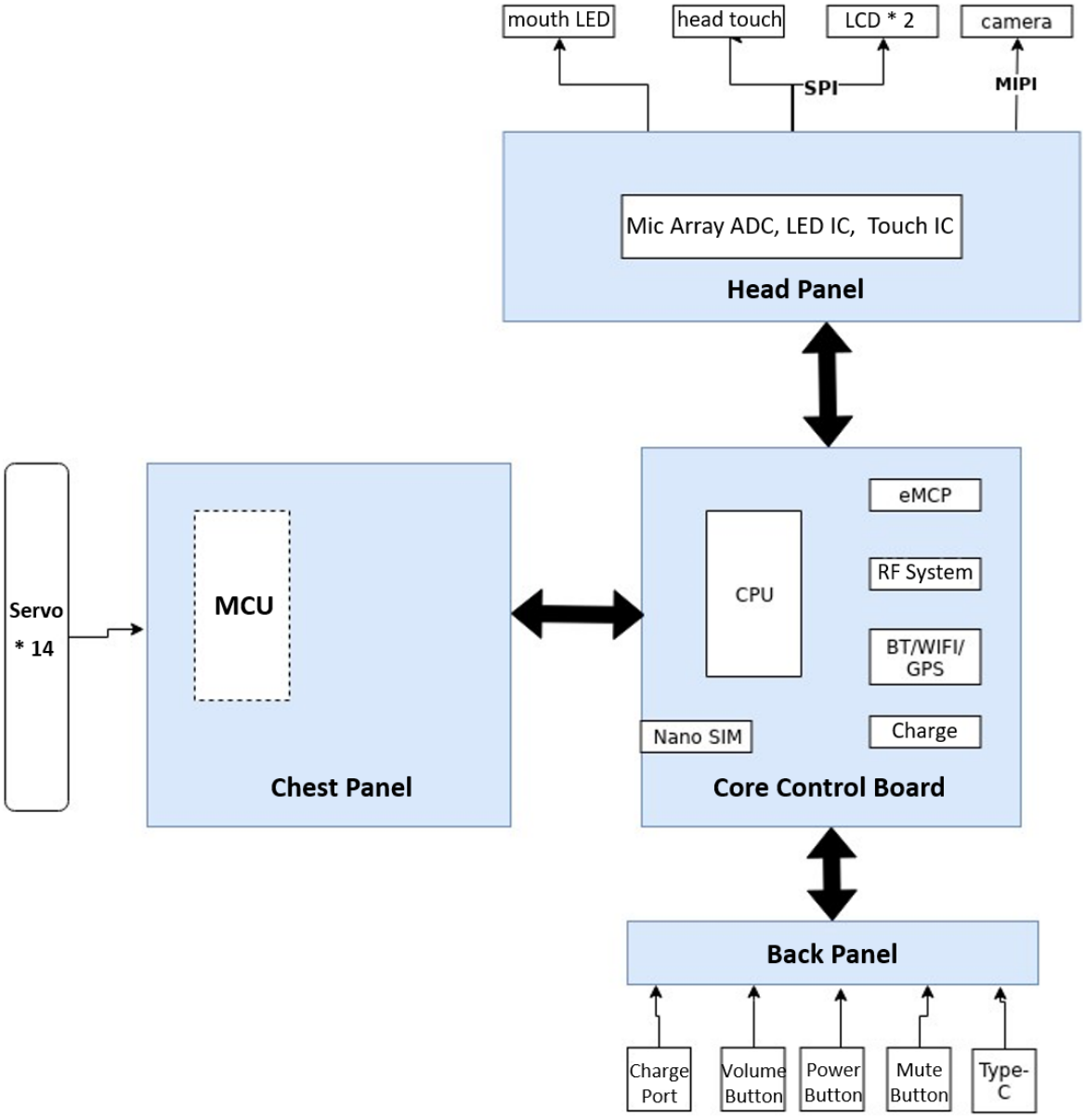
Power Saving Control

To implement power saving on the robot, please refer to the diagram below to understand hardware structure on the robot:



1. There are 4 sets of chipboard embedded in the robot: Core Control Board, Chest Panel, Head Panel and Back Panel; Core Control Board and Chest Panel is at the chest of the robot; Android 7.0 OS is operating by the CPU in the Core Control

Board. The MCU of Chest Panel controls all the servo motors; Head Panel is connected with mouth LED, LCD display, Mic Array and Camera; Buttons like volume, power is connected to Back Panel.

2. Power saving mechanism for Camera, Wi-Fi, BT, GPS and other common components of Cellphone could refer to standard Android OS power saving strategy.

3. Power saving for Servo motors, IR sensor, Chest Panel, Mic Array, Mouth LED and 2 LCD displays need use interfaces provided by UBTECH to implement.

Power Saving for Servo Motors

Lock and unlock servo motors:

In other explanations, a user can twist the unlocked servo motors but cannot twist the locked servo motors. Obviously, if the user feel a joint of robot is loose, so the related servo motors are unlocked, if it feels tight, so the servo motors are locked. When a servo motor is unlocked, it will consume less power than locked status, and it is still possible to read and set angle of the servo motor. Please be noticed the difference: when a servo motor is totally power off, it is not accessible to read and set angle.

To check whether a servo motor is locked or not, there are some functions:

```
/**
* lockMotor: if a motor is moving or unlocked, use this function to lock the motor
*
* @param motorIds
* @param listener
* @param priority
*/
public void lockMotor(@NonNull final List<Integer> motorIds, Priority priority,
@Nullable final ResponseListener<Boolean> listener);
/**
* unlockMotor: if a motor is moving, this function will stop the movement, and the
motor is unlocked, if the motor is locked, this function will unlock it
*
* @param motorIds
* @param priority
* @param listener ResponseListener
*/
public void unlockMotor(@NonNull final List<Integer> motorIds, Priority
priority,@Nullable final ResponseListener<Boolean> listener) ;
```

Power on and off servo motors:

When a servo motor is power off, it will not receive any command until power on.

Once power off, a servo motor takes 2~3 seconds to power on again.

To check whether a servo motor is power on or off, there are some functions:

```
public class MotorApi {

/**

* Servo motor powerOff

*/public void powerOff() ;

/**

* Servo motor powerOn (with Listener)

*

* @param listener listener of call

*/


public void powerOn(@Nullable final ResponseListener<Boolean> listener);

/**

* Once being power off, a servo motor takes 2~3 seconds to power on again.

*/

public void powerOn();

}
```

Power on and off IR sensor:

IR sensor is connected to Chest Panel, please refer to the functions below:

```
public class IrApi {

/**

* IR powerOn

*

* @return boolean

*/

public boolean powerOn();

/**

* IR powerOff

*

* @return boolean

*/

public boolean powerOff();

}
```

Power on and off for Chest Panel:

Set the power on and off for Chest Panel will also affect on MCU/Servo motors

and IR sensor; If Chest Panel is power off, all the connected components will be power off.

If Chest Panel is set to power off, servos will be power off at the first, then IR senor, Chest Panel will be the last to power off. On reverse, it is the sequence of power on, no need to call MotorApi.powerOn, IrApi.powerOn functions.

Once Chest Panel is power off, Chest Panel, servos and IR sensor interface will be all invalid, there are functions to power on and off Chest Panel:

```
public class SysApi {
/**
* Shut down Chest Panel—Chest Panel power off
*/
public void shutdown();
/**
* Restart Chest Panel: Chest Panel will be started after at least 6000 ms
*
* @param listener {@link ResponseListener}
*/
public void startup(@NonNull final ResponseListener<Void> listener) ;
}
```

Mouth LED

The mouth LED is a LED light located at the bottom of the robot’ s head as the “mouth”. Usually , the duration of LED light on could be controlled by APIs and the LED will be power off after the duration, please refer to these functions:

```
public class MouthLedApi {

/**

* LED light on—Always on

*/

@Override public void turnOn() ;

/**

* LED light off

*/

@Override public void turnOff() ;

/**

* LED light on and set color and duration

*

* @param color

* @param duration, -1 for Always on

*/

@Override public void startNormalModel(int color, int duration);

}
```

LCD display

Two of 240*240 px LCD displays are on the head of the robot as the “eyes”, these functions are used for controlling the back light on/off and brightness of the LCD displays:

```
public class BackLightApi {
/**
* Power on back light
*/
public void powerOn();
/**
* Power off back light
*/
public void powerOff();
/**
* Set backlight brightness
*
* @param brightness rang of [0, 16]
*/
public void setBrightness(@IntRange(from = 0, to = 16) int brightness);
}
```